

INSTRUCTIONS: Determine whether each equation is linear. If it is, solve for its root(s); if not, just write "Not Linear."

EXAMPLE:

a) $4x - 5 = 3(x + 2)$

Distribute: $4x - 5 = 3x + 6 \rightarrow x = 11$. Only x -to-the-first-power appears, so this is LINEAR. Root: $x = 11$.

b) $x^2 + 2x = 8$

This has an x^2 term — it cannot be written as $ax + b = 0$. NOT LINEAR (no need to solve further).

c) $3/(x - 2) + 2 = 5/(x - 2)$

Multiply both sides by $(x - 2)$: $3 + 2(x - 2) = 5 \rightarrow 2x - 1 = 5 \rightarrow x = 3$. Since $x = 3$ does not make the denominator zero, it is a valid root. LINEAR. Root: $x = 3$.

I. LINEAR EQUATIONS — RECOGNIZING AND SOLVING

1. $x - 9 = 15$

2. $3x + 7 = 2x - 5$

3. $5(x - 3) = 2(x + 6)$

4. $6(2x - 1) - 4x = 3(x + 5)$

5. $7 - 3(x + 2) = 4x - 1$

6. $2(3x - 1) = 4x + 10$

7. $4x - 7 = 4x + 2$

8. $x + 5 = x + 5$

9. $x^2 - 5x = 6$

10. $2x^2 + 3x = 0$

11. $x^3 - x = 0$

12. $x(x + 3) = 10$

II. FRACTIONAL AND DECIMAL EQUATIONS

13. $(x + 3)/5 + (x - 1)/2 = 5$

14. $(2x + 1)/3 - (x - 4)/5 = 3$

15. $(x - 2)/4 + (2x + 1)/6 = 2$

16. $(3x + 2)/5 - (x - 1)/3 = 1$

17. $0.3x/5 + 0.4x = 2.3$

18. $0.5(x + 4) - 0.25x = 3.5$

19. $0.6(x - 2) = 0.4x + 1.6$

20. $(2/3)(x - 6) - (1/4)(x + 8) = -1$

21. $3(x + 1)/4 - 2(x - 1)/3 = 1/6$

22. $x^2/x - 3 = 5$

23. $\frac{x}{2} + \frac{x}{3} - \frac{x}{6} = 4$

III. EQUATIONS WITH RATIONAL EXPRESSIONS (VARIABLE IN THE DENOMINATOR)

24. $\frac{3}{(x-2)} + 2 = \frac{5}{(x-2)}$

25. $\frac{5}{(2x-3)} = \frac{3}{(x+1)}$

26. $\frac{x}{(x-1)} + 1 = \frac{2}{(x-1)}$

27. $\frac{4}{(x+3)} - 1 = \frac{1}{(x+3)}$

28. $\frac{6}{(x^2-1)} = \frac{2}{(x-1)} - \frac{1}{(x+1)}$

29. $\frac{2}{(x+4)} + \frac{1}{(x-4)} = \frac{3}{(x^2-16)}$

30. $\frac{3}{(x+2)} + \frac{2}{(x-2)} = \frac{5}{(x^2-4)}$

31. $\frac{(x+1)}{(x-3)} - 2 = \frac{4}{(x-3)}$

32. $\frac{1}{(x-1)} + \frac{1}{(x+1)} = \frac{2}{(x^2-1)}$

33. $\frac{x}{(x+1)} - \frac{x}{(x-1)} = \frac{-2}{(x^2-1)}$

34. $\frac{2}{(x-3)} + \frac{3}{(x+3)} = \frac{5x}{(x^2-9)}$

35. $\frac{(2x)}{(x+1)} - 1 = \frac{(x-1)}{(x+1)}$

36. $\frac{(x+2)}{(x-2)} = \frac{(x-2)}{(x+2)}$

37. $\frac{3}{x} - x = 2$

38. $\frac{2}{(x-1)} + x = 4$

39. $\frac{x^2}{(x+2)} = x - 2$

IV. RADICAL EQUATIONS

40. $\sqrt{2x+3} = 5$

41. $4 - \sqrt{2x-3} = 1$

42. $5 + \sqrt{3x-5} = 9$

43. $2\sqrt{x-1} - 4 = 0$

44. $3\sqrt{x} + 2 = 11$

45. $\sqrt{2x-1} = x - 2$

46. $\sqrt{5x-4} = x - 2$

47. $\sqrt{4x+5} = 2x - 5$

48. $\sqrt{6x+1} = 2x - 1$

49. $\sqrt{x+10} = x - 2$

50. $\sqrt{2x+7} = x + 2$

51. $\sqrt{x+7} - \sqrt{x} = 1$

52. $\sqrt{3x+1} - \sqrt{x-1} = 2$

53. $\sqrt{x+3} + \sqrt{x} = 3$

54. $\sqrt{x^2 + 16} = x + 2$

Mastery Checklist — Situations You Should Recognize on Sight

Section I — Basic Linear Equations

- Already isolated or one step away (I.A)
- Needs distribution before solving (I.B)
- Variables end up on both sides after distributing (I.C)
- Simplifies to a false statement → No Solution (I.D)
- Simplifies to a true statement for every x → Infinitely Many Solutions (I.D)
- Contains x^2 , x^3 , or a product that expands into x^2 → Not Linear (I.E)

Section II — Fractional and Decimal Equations

- Numeric fractions — clear with the LCD (II.A)
- Decimal coefficients (II.B)
- Distribution combined with fractions (II.C)
- Looks non-linear (like x^2/x) but simplifies first (II.D)
- Several fraction terms to combine over one LCD (II.E)

Section III — Variable in the Denominator

- Same denominator on both sides (III.A)
- Clears directly to an ordinary linear equation (III.B)
- LCD is a difference of squares (III.C)
- Only candidate root turns out extraneous → No Solution (III.D)
- All candidate roots cancel out → No Solution (III.E)
- Result is an identity → Infinitely Many Solutions, with a restriction (III.F)
- Cross-multiplying looks like it creates x^2 , but the x^2 cancels → still Linear (III.G)
- Clearing denominators creates a genuine quadratic that does not cancel → Not Linear (III.H)
- x^2 cancels but what's left is a false numerical statement → No Solution (III.I)

Section IV — Radical Equations

- Single radical — isolate, then square once (IV.A)
- Radical with a coefficient in front, like $3\sqrt{x}$ (IV.B)
- Squaring creates an x^2 that does not cancel → Not Linear, check for extraneous roots (IV.C)
- Two radicals that reduce to a linear equation (IV.D)
- Two radicals that reduce to a quadratic → Not Linear (IV.E)
- Two radicals in sum form (IV.F)
- x^2 under the radical cancels the squared side → still Linear (IV.G)

Answer Key

1. // already close to isolated

Linear — $x = 24$

2. // isolate x terms on one side, constants on the other

Linear — $x = -12$

3. // distribute, then isolate x

Linear — $x = 9$

4. // distribute, then isolate x

Linear — $x = 21/5$

5. // distribute the negative signs carefully, then collect terms

Linear — $x = 2/7$

6. // distribute, then isolate x

Linear — $x = 6$

7. // the x -terms cancel out, leaving a false numerical statement

Linear — No Solution

8. // both sides are identical, true for every x

Linear — Infinitely Many Solutions

9. // has an x^2 term once simplified

Not Linear

10. // has an x^2 term

Not Linear

11. // has an x^3 term

Not Linear

12. // expand the product first — it introduces an x^2 term

Not Linear

13. // clear fractions with the LCD, then solve

Linear — $x = 7$

14. // clear fractions with the LCD, then solve

Linear — $x = 4$

15. // clear denominators using the LCD, then solve

Linear — $x = 4$

16. // clear denominators using the LCD, then solve

Linear — $x = 1$

17. // combine the x -terms first, then divide

Linear — $x = 5$

18. // distribute, then isolate x

Linear — $x = 6$

19. // distribute, then isolate x

Linear — $x = 14$

20. // distribute each fraction, then collect like terms

Linear — $x = 12$

21. // clear fractions with the LCD, then solve

Linear — $x = -15$

22. // x^2/x simplifies to x ($x \neq 0$) before you solve

Linear ($x \neq 0$) — $x = 8$

23. // clear denominators using the LCD (6), then solve

Linear — $x = 6$

24. // same denominator on both sides — combine directly, then check $x \neq 2$

Linear — $x = 3$

25. // cross-multiply, then solve; check both denominators are nonzero

Linear — $x = 14$

26. // multiply both sides by $(x - 1)$, solve, then check the restriction

Linear — $x = 3/2$

27. // multiply both sides by $(x + 3)$, solve, then check the restriction

Linear — $x = 0$

28. // combine the right side into a single fraction first, then compare numerators

Linear — $x = 3$

29. // note $x^2 - 16 = (x+4)(x-4)$; multiply through by this LCD

Linear — $x = 7/3$

30. // multiply by the LCD $(x+2)(x-2)$, then solve

Linear — $x = 7/5$

31. // multiply both sides by $(x - 3)$, solve, then check against the restriction

Linear — No Solution ($x = 3$ is extraneous)

32. // multiply by the LCD $(x-1)(x+1)$ — the x^2 terms cancel, but check restrictions

Linear — No Solution ($x = 1$ is extraneous)

33. // multiply by the LCD $(x-1)(x+1)$ — the x^2 terms cancel, but check restrictions

Linear — No Solution ($x = 1$ is extraneous)

34. // multiply by the LCD — the x -terms cancel, leaving a false statement

Linear — No Solution

35. // multiply by $(x + 1)$ — both sides become identical

Linear — Infinitely Many Solutions ($x \neq -1$)

36. // cross-multiply and expand — the x^2 terms cancel out

Linear — $x = 0$

37. // multiply both sides by x — this introduces an x^2 term

Not Linear (bonus: $x = -3$ and $x = 1$, both check)

38. // multiply both sides by $(x - 1)$ — this introduces an x^2 term

Not Linear (bonus: $x = 2$ and $x = 3$, both check)

39. // multiply both sides by $(x + 2)$ — the x^2 terms cancel, leaving a false statement

Linear — No Solution

40. // square both sides once the radical is isolated

Linear — $x = 11$

41. // isolate the radical, then square

Linear — $x = 6$

42. // isolate the radical, then square

Linear — $x = 7$

43. // isolate the radical, then square

Linear — $x = 5$

44. // isolate the radical, then square

Linear — $x = 9$

45. // squaring $(x - 2)$ introduces an x^2 term

Not Linear (bonus: $x = 5$; $x = 1$ is extraneous)

46. // squaring $(x - 2)$ introduces an x^2 term

Not Linear (bonus: $x = 8$; $x = 1$ is extraneous)

47. // squaring $(2x - 5)$ introduces an x^2 term

Not Linear (bonus: $x = 5$; $x = 1$ is extraneous)

48. // squaring $(2x - 1)$ introduces an x^2 term

Not Linear (bonus: $x = 5/2$; $x = 0$ is extraneous)

49. // squaring $(x - 2)$ introduces an x^2 term

Not Linear (bonus: $x = 6$; $x = -1$ is extraneous)

50. // squaring $(x + 2)$ introduces an x^2 term

Not Linear (bonus: $x = 1$; $x = -3$ is extraneous)

51. // isolate one radical, square, then isolate and square again

Linear — $x = 9$

52. // isolating and squaring leaves a squared term in the intermediate variable

Not Linear (bonus: $x = 1$ and $x = 5$, both check)

53. // isolate one radical, square — the x terms cancel, leaving a simple radical equation

Linear — $x = 1$

54. // the x^2 under the radical cancels with the x^2 from squaring $(x + 2)$

Linear — $x = 3$